

MANTHAN GARG

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PROFILE

Computer Science Engineering student at Chitkara University focused on cybersecurity and endpoint threat detection. Hands-on experience with YARA-based static analysis, threat intelligence APIs, MITRE ATT&CK mapping, and system telemetry. Foundations in Python, C++, Linux, network security, cryptography, and operating systems.

EDUCATION

Chitkara University

2024-2028

B.E. Computer Science Engineering | CGPA: 8.34

Relevant Coursework: Network Security, Cryptography, Operating Systems, Data Structures & Algorithms, OOP (C++)

PROJECTS

SENTRA CORE V2 | Endpoint Security and Threat Intelligence Platform

June 2026 - Present

- Built a Python/FastAPI endpoint security platform combining YARA signature matching, Shannon entropy analysis, heuristic risk scoring, and keyword-based MITRE ATT&CK technique mapping.
- Integrated VirusTotal, MalwareBazaar, AlienVault OTX, AbuseIPDB, and URLhaus for SHA-256 file-hash and IP reputation checks, with cached verdicts and Windows Authenticode fallback.
- Streamed CPU, memory, process, and network telemetry to a React dashboard via 1-second WebSocket updates; added SQLite scan history, PDF audit reports, and scheduled scans.

NeuroSole | Smart Healthcare Wearable (Patent Published) | neurosole.vercel.app

Feb 2026 - May 2026

- Developed an embedded data acquisition and sensor-fusion pipeline using FSR, MPU6050 IMU, and EMG sensors; applied digital signal processing to classify plantar-pressure patterns.
- Implemented asynchronous data ingestion and signal filtering between microcontroller firmware and a cloud analytics dashboard for continuous biometric telemetry.
- Received patent publication for the hardware-software integration combining real-time sensor fusion with a remote healthcare platform.

NervoStep-ML | Multimodal ML Research Prototype

April 2026 - June 2026

- Implemented GRU recurrent networks and 1D convolutional neural networks (CNNs) in TensorFlow/Keras for multimodal time-series fusion and three-class classification.
- Built NumPy preprocessing and subject-aware ECG train/validation splits; applied dropout regularization and early stopping, with SHAP feature-attribution heatmaps on simulated inputs.
- Built a Streamlit/Plotly dashboard with SVG gait animations and time-series plots using simulated scenario data.

TECHNICAL SKILLS

Cybersecurity: YARA, Entropy Analysis, Heuristic Detection, Network Security, Cryptography, MITRE ATT&CK

Programming: C++, Java, Python, JavaScript, Bash, HTML

Systems and IoT: Linux (Arch), Windows System APIs, Embedded Systems, IoT Architecture, Sensor Fusion

Development: Git, Node.js, REST APIs, FastAPI

ML and Data: TensorFlow, Keras, SHAP, Streamlit, Plotly, NumPy

Core Concepts: Data Structures & Algorithms, OOP, Design Patterns, Concurrency, OS Internals, Signal Processing

CERTIFICATIONS

Introduction to Cybersecurity | Cisco Networking Academy

Cybersecurity Fundamentals | IBM SkillsBuild

Introduction to Modern AI | Cisco Networking Academy

Introduction to Data Science | Cisco Networking Academy

Introduction to IoT and Digital Transformation | Cisco Networking Academy

LEADERSHIP AND ACHIEVEMENTS

- **Vice Chair, IEEE Student Branch, Chitkara University:** Led technical events and cross-chapter collaboration.
- **Team Member, IEEE Delhi SAC and IOC Team:** Contributed to regional student activity planning and execution.
- **IEEE DSSYWLC 2025:** Led sponsorship outreach and designed the official sponsorship brochure.
- **Top 5 Team - Buildathon, Emergence'26**